

✓ 10 — Report generation

I gather generated figures, tables, and key metrics into a structured project summary.

✓ V3 orientation-label interpretation

I treat continuous visual-feature decoding as the primary analysis. Hard orientation labels from natural movies can be imbalanced and ambiguous, so I add a confidence-aware orientation label that marks low-selectivity frames as ambiguous. Orientation-class decoding is reported as secondary/exploratory with majority-class and shuffled-label baselines.

```
from pathlib import Path
import sys

ROOT = Path.cwd()
if ROOT.name == "notebooks":
    ROOT = ROOT.parent
if str(ROOT / "src") not in sys.path:
    sys.path.insert(0, str(ROOT / "src"))

from v1_manifold.config import load_config, get_paths, set_global_seed
from v1_manifold.visualization import set_publication_style, save_figure
from v1_manifold.utils import save_table

cfg = load_config(ROOT / "configs" / "default.yaml")
cfg["paths"]["root"] = str(ROOT)
paths = get_paths(cfg)
set_global_seed(cfg["project"]["random_seed"])
set_publication_style()
print(f"Project root: {ROOT}")
```

Project root: c:\Users\Peter\Documents\projects\NeuroAI\latent-manifold-v1-na

```
from v1_manifold.reporting import collect_report_assets, write_markdown_report

assets = collect_report_assets(paths.figures_dir, paths.tables_dir)
summary = ""
I analysed Allen Brain Observatory V1 calcium-imaging responses during natural
The workflow produced validated trial tensors, real visual features extracted
Allen natural movie frames, population-level exploratory analyses, low-dimensi
representations, baseline decoding models, CEBRA/dRNN advanced models, geometri
event-triggered interpretability analyses, and manuscript-ready report assets.
```

The main stimulus-decoding result is that held-out-repeat V1 population activity is best predicted by continuous image-derived visual statistics, including RMS contrast, luminance, orientation selectivity, spatial-frequency centroid, and total spectral power. We use dominant-orientation classification as secondary/exploratory because natural image labels are imbalanced and temporally autocorrelated.

The latent-representation analyses show that PCA, UMAP, ISOMAP, and CEBRA provide low-dimensional descriptions of the same V1 population trajectory. Under block cross-validation, latent embeddings showed limited real-feature predictivity, but these embedding-feature regressions as descriptive rather than as strong predictive. The dRNN next-state model captured temporal structure in the learned CEBRA trajectory. Performance should be compared against the persistence baseline rather than in

The geometry and interpretability analyses showed heavy-tailed speed, curvature events in the CEBRA trajectory. These high-geometry events often coincide with interpretable natural-movie frames, but geometry-stimulus correlations were modest. I interpret them as descriptive alignment between latent neural geometry and stimulus, not as causal evidence.

```
"""
report_path = paths.root / "reports" / "html" / "project_summary.md"
write_markdown_report(report_path, "V1 natural movie manifold project summary")
print(f"Saved report summary to {report_path}")
print(f"Figures: {len(assets['figures'])}, tables: {len(assets['tables'])}")
```

Saved report summary to C:\Users\Peter\Documents\projects\NeuroAI\latent-manifold\reports\html\project_summary.md
Figures: 66, tables: 56

```
import pandas as pd

for table in assets["tables"][:5]:
    print("\n", table.name)
    try:
        display(pd.read_csv(table).head())
    except Exception as exc:
        print(exc)
```

00_workflow_plan.csv

	stage	notebook	main_output
0	1	data acquisition	Allen experiment catalog
1	2	loading and validation	shape/schema checks
2	3	preprocessing	trial tensors
3	4	EDA	population activity figures
4	5	latent representations	PCA/UMAP/ISOMAP embeddings

01_available_v1_natural_movie_cpe_lines.csv

01_available_v1_natural_movie_cre_lines.csv

	cre_line	imaging_depth	session_type	n_experiments
0	Cux2-CreERT2	175	three_session_A	8
1	Cux2-CreERT2	175	three_session_B	8
2	Cux2-CreERT2	175	three_session_C	7
3	Cux2-CreERT2	175	three_session_C2	1
4	Cux2-CreERT2	275	three_session_A	8

01_selected_experiments_by_cre_depth_layer.csv

	cre_line	imaging_depth	putative_layer	n_experiments
0	Cux2-CreERT2	175	L2/3	3

01_selected_experiments_by_cre_line.csv

	cre_line	n_experiments
0	Cux2-CreERT2	3

02_validation_experiment_500855614.csv

	object	n_cells	n_timepoints	n_presentations	n_cells_metadata
0	dff_traces	164.0	105959.0	NaN	NaN
1	stimulus_table	NaN	NaN	9000.0	NaN
2	cell_metadata	NaN	NaN	NaN	164.0

Final interpretation template

After I run the full pipeline, I complete this section with the strongest empirical pattern: which embedding has the lowest tangling, which representation best predicts continuous real movie features, and whether any classification performance exceeds the majority-class and shuffled-label baselines. I report fine orientation decoding only as an exploratory result if class imbalance remains severe. I also distinguish ordinary frame-wise neural--stimulus correlations from circular-shift-null-supported relationships because adjacent natural movie frames are not independent.

